How-To Library

Informatica

Configuring Auto Curation Parameters to Infer Foreign Keys in Informatica Developer

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Abstract

When you run an enterprise discovery profile in Informatica Developer, you can discover column profile statistics, data domains, primary keys, and foreign keys in a large number of data sources. A large number of keys and relationships might be inferred. You can approve, reject, or reset the inferred data domains, primary keys, or foreign keys. This article describes how you can configure the auto curation parameters to infer primary key and foreign key relationships without manual intervention. The auto curation parameters are user-defined custom attributes that you can configure to identify the data relationships based on certain conditions.

Supported Versions

- Data Quality 10.0

Table of Contents

Overview. 2
Foreign Key Configuration File. 3
Data Overlap Match. 3
Name Match. 3
Relationship Type Match. 3
Data Type Match. 4
Attributes for Auto Curation Parameters. 4
Business Example. 4
Creating Enterprise Discovery Profile to Discover Foreign Keys. 6

Overview

You can create and run an enterprise discovery profile to discover primary key and foreign key relationships between multiple data objects. You can infer the primary keys and foreign keys based on the default values, or you can set the primary key and foriegn key inference options in Informatica Developer.

When you perform foreign key discovery, most of the possible primary key and foreign key relationships between the data objects in the enterprise discovery profile appear. You can curate the inferred relationships with the approve, reject, or reset options.

When the discovery results include a large number of primary key and foreign key relationships, you might find it tedious to identify the critical data relationships among hundreds of data relationships. You might also find it difficult to curate the relationships based on certain conditions, such as data match or data type. To resolve this issue, you can configure the auto curation parameters and run the enterprise discovery profile.

If the data sources have multiple candidate foreign keys and you want to provide rules to choose a candidate foreign key, you can perform the following actions:

- Configure the **Max Foreign Keys between data objects** and **Minimum conformance percent** options in the enterprise discovery profile wizard.
- Configure the weights and scores for the auto curation parameters in the `ForeignKeyConfig.xml` file.

Foreign Key Configuration File

An administrator can edit and save the foreign key configuration file. Configure the auto curation parameters in the foreign key configuration file. The algorithm infers the primary key and foreign key relationships between multiple data objects based on the auto curation parameters.

The foreign key configuration file, `ForeignKeyConfig.xml`, is available in the following directory:

```
<Informatica installation directory>\services\DataIntegrationService\modules\ProfilingService
```

The auto curation parameters are data overlap match, column name match, relationship type match, and data type match.

Data Overlap Match

Data overlap match is the estimated overlap of values between the primary key and foreign keys. You can set the overlap match in the enterprise discovery profile wizard with the **Minimum conformance percent** option. By default, the **Minimum conformance percent** option is set to 90.

If the data overlap match does not meet the minimum conformance percentage, the foreign key is not considered for auto curation. When the minimum conformance for the data overlap match is met, the remaining parameters are used to compute the adjusted score.

Name Match

The name match parameter is an optional parameter. It uses the Edit Distance algorithm to determine how closely the names of primary and foreign key columns match and sets the score between 0 and 1. Set the name match weight to 0 if you do not want to use this parameter to determine primary key and foreign key relationship.

Relationship Type Match

The relationship type match determines the type of relationship between the primary key and foreign key columns and assigns a fixed score between 0 and 1. The relationship type match is computed based on the column type of the foreign key column.

The following relationship type matches can be set in the `ForeignKeyConfig.xml` file:

- Primary key-foreign key relationship where the foreign key column is a non-key column. The default for this relationship type match is 1. You can find this relationship in many data sources.
- Primary key-primary key relationship where the foreign key column is a primary key column. The default for this relationship type is 0.25. You rarely find this relationship type as it represents a table that has been partitioned vertically.
- Primary key-primary key sequence relationship where the foreign key column is a primary key column and the data type of the column is a sequence data type. For example, OrderID column in a Order table has a sequence data type. The default for this relationship type is zero because the sequence keys might cause multiple false positive foreign keys, which the primary key-primary key algorithm tries to avoid. You can set the relationship type match to a higher score if the data source is known to contain a few sequence data types.

Data Type Match

The data type match compares the data types of the primary key columns to the foreign key columns and assigns a fixed conformance score which is based on how close the data types of the columns match.

The following table lists the fixed data type match scores for different combinations of primary key and foreign keys:

	Numeric Foreign Key	**Date Foreign Key**	**String Foreign Key**
Numeric Primary Key	1.0	0.5	0.0
Date Primary Key	0.5	1.0	0.5
String Primary Key	0.0	0.0	1.0

You can change the default data type match scores if required.

Attributes for Auto Curation Parameters

You can use default weights and set the scores for the auto curation parameters in the `ForeignKeyConfig.xml` file. After you set the scores, the weighted score for the parameters and the final score are computed by the Data Integration Service.

The auto curation parameters have the following attributes:

- Score. The partial conformance percentage for an auto curation parameter. You can configure this attribute based on your requirements.
- Weight. The weight that you can set for an auto curation parameter. You can use the default weights or set this attribute as per your enterprise requirements when you configure the `ForeignKeyConfig.xml` file for the first time. If the enterprise requirements change over time, you can change the weight for the auto curation parameters.
  If you do not set the name match parameter, the system automatically normalizes the sum of weights to 100%.
- Weighted score. The weighted score for an auto curation parameter is calculated as *weight * score*.

The conformance score is calculated after the weighted score for each parameter is computed. The conformance score is calculated as *(Sum of the weighted scores / Sum of the weights) * 100%*. The weights need not sum up to 100% because the Data Integration Service automatically normalizes the weights to 100% when computing the weighted score.

Business Example

You are a data quality developer in an organization. Your organization has acquired a startup. Before you merge the data from both the organizations, you want to analyze the source data and metadata in the database of the startup company. Your organization recognizes that matching data is better than relying on names so you want to make it a priority in all your tasks. First, you want to identify primary key and foreign key relationships in the tables in the startup database.

To identify the primary key and foreign key relationships between the Employee table and Payroll table, perform the following steps in Informatica Developer:

1. Create relational data objects for the Employee table and Payroll table.

2. Create an enterprise discovery profile on the tables and configure the following options for foreign key profiling in the profile wizard.
   - Max Foreign Keys between data objects = 2
   - Minimum conformance percent = 96
3. To implement your organization's policy for making data matching a priority, increase the **Data Overlap Match** weight from the default value to a higher value to make the data overlap match relatively important with respect to the rest of the parameters. For example, you can change the **Data Overlap Match** weight from 0.8 to 1.1. Note that the value of the data overlap match weight is to be considered carefully.
   In the `ForeignKeyConfig.xml` file, configure the following weight and scores for the auto curation parameters:

Parameter	Weight	Score1	Score2	Score3
Data overlap match	1.1	0.96	0.96	0.98
Name match	0.3	0.49	0.55	0.49
Relationship match	0.1	1.00	1.00	1.00
Data Type match	0.1	1.00	1.00	1.00

4. Run the profile.
5. Refresh the Model repository.

   Note: You need to perform this action as the import of metadata for external connections happens in the Model repository. You need to refresh the Model Repository Service so that the Developer tool reflects the changes to the Model repository.
6. View the profile results.

The foreign keys ForeignKey1, ForeignKey2, and ForeignKey3 are inferred which differ in the overlap data and column names.

The following table lists the sample scores, weights, calculated weighted scores, and the conformance score for each inferred foreign key:

Parameter	Weight	ForeignKey1		ForeignKey2		ForeignKey3	
		Score	Weighted Score	Score	Weighted Score	Score	Weighted Score
Data overlap match	1.1	0.96	1.06	0.96	1.06	0.98	1.08
Name match	0.3	0.49	0.15	0.55	0.17	0.49	0.15
Relationship match	0.1	1.00	0.10	1.00	0.10	1.00	0.10
Data Type match	0.1	1.00	0.10	1.00	0.10	1.00	0.10
Total			1.40		1.42		1.43
Conformance Score		87.69%		88.81%		89.06%	

The algorithm retains ForeignKey2 and ForeignKey3 foreign keys and discards the ForeignKey1 foreign key because the **Max Foreign Keys between data objects** option is set to 2.

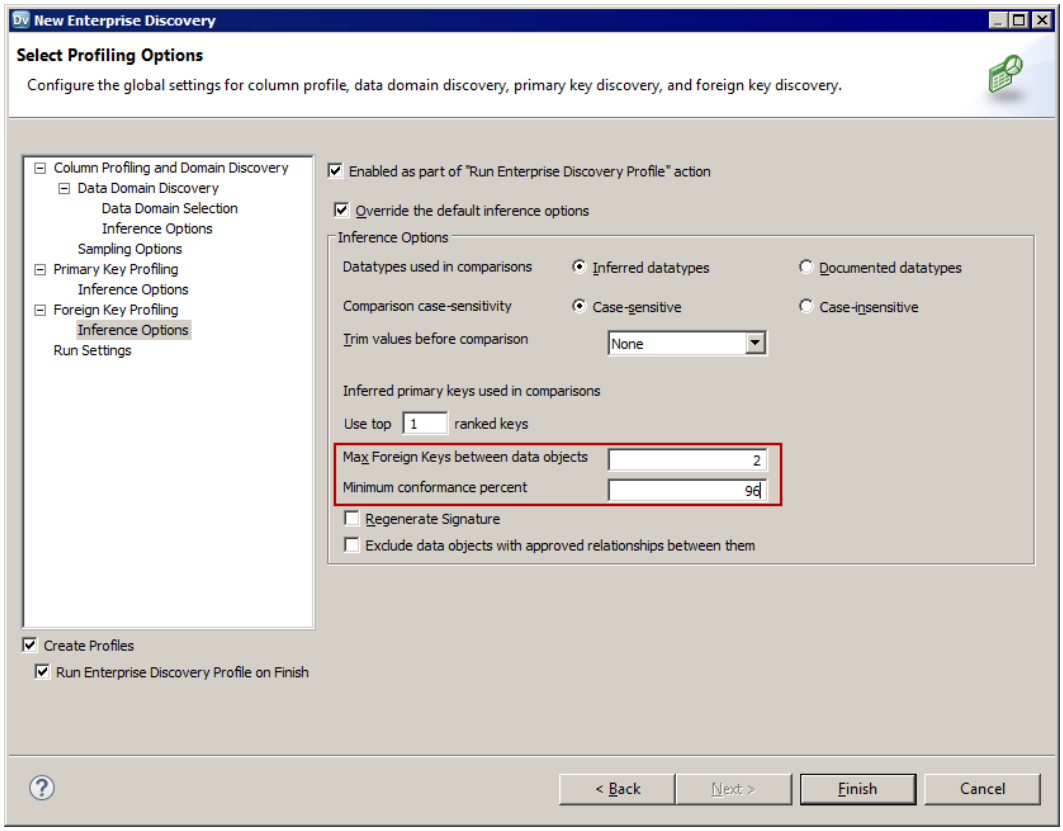
Creating Enterprise Discovery Profile to Discover Foreign Keys

You can identify the primary key and foreign key relationships between multiple tables after you set the primary key inference options, foreign key inference options, and set the auto curation parameters in the `ForeignKeyConfig.xml` file.

For example, to identify the primary key and foreign key relationships between the Employee table and Payroll table, perform the following steps:

1. Create relational data objects for the Employee and Payroll tables.
2. In the **Object Explorer** view, select the Employee and Payroll data objects.
3. Click **File** > **New** > **Profile** to open the profile wizard.
4. Select **Enterprise Discovery Profile** and click **Next**.
5. Enter **PKFK1** as the name for the profile and verify the project location. Click **Next**.
6. Verify that the Employee and Payroll data objects appear within the **Data Objects** section. Click **Next**.
7. In the **Add Resources to Profile Definition** screen, click **Next**.
8. In the **Select Profiling Options** screen, click **Foreign Key Profiling** > **Inference Options** and set the following options:
   - Select **Override the default inference options**.
   - Configure the **Max Foreign Keys between data objects** option.
   - Configure the **Minimum conformance percent** option.

   The following image shows the foreign key inference options in the enterprise discovery profile wizard:



9. Click **Finish**.

10. Open the `ForeignKeyConfig.xml` file. The file is located in the `<Informatica installation directory>\services\DataIntegrationService\modules\ProfilingService` directory.
11. In the `ForeignKeyConfig.xml` file, enter the scores for data overlap match, name match, relationship type match, and data type match.

    The following image shows a sample `ForeignKeyConfig.xml` file with default auto curation parameters:

```
<?xml version="1.0" encoding="UTF-8"?>
<!--
This XML contains a list of configurations/knobs that would be
used in the FK inference algorithm to boost or penalize certain
foreign keys to minimize the false positives.
-->

<ForeignKeyInferenceConfig>

    <NameMatch weight="0.3"/>

    <RelationshipTypeMatch weight="0.1">
        <score match="pkfk" value="1"/>
        <score match="pkpk" value="0.25"/>
        <score match="pkpk_seq" value="0"/>
    </RelationshipTypeMatch>

    <DatatypeMatch weight="0.1">
         <score match="Numeric;Numeric" value="1"/>
         <score match="Numeric;Date" value="0.5"/>
         <score match="Numeric;String" value="0"/>

         <score match="Date;Numeric" value="0.5"/>
         <score match="Date;Date" value="1"/>
         <score match="Date;String" value="0.5"/>

         <score match="String;Numeric" value="0"/>
         <score match="String;Date" value="0"/>
         <score match="String;String" value="1"/>

    </DatatypeMatch>

    <DataOverlapMatch weight="0.8"/>

</ForeignKeyInferenceConfig>
```


12. Save the file.
13. In the Developer tool, run the enterprise discovery profile.
14. Refresh Model Repository.
15. View profile results.
16. To view the consolidated view of foreign key relationships across the data objects, click **Relationships** > **Foreign Key Profile**.

Author

Lavanya S

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